

Exemplar: how a heterotroph rescues *Prochlorococcus* from nitrogen starvation

KG-driven re-analysis reproduces and extends Weissberg et al. 2025 — a 90-day axenic-vs-coculture multi-omics time course

The system

Prochlorococcus MED4 in N-poor seawater (PRO99-lowN), continuous light, 24 °C — grown two ways:

Axenic

dies in ~2 weeks

+ *Alteromonas*

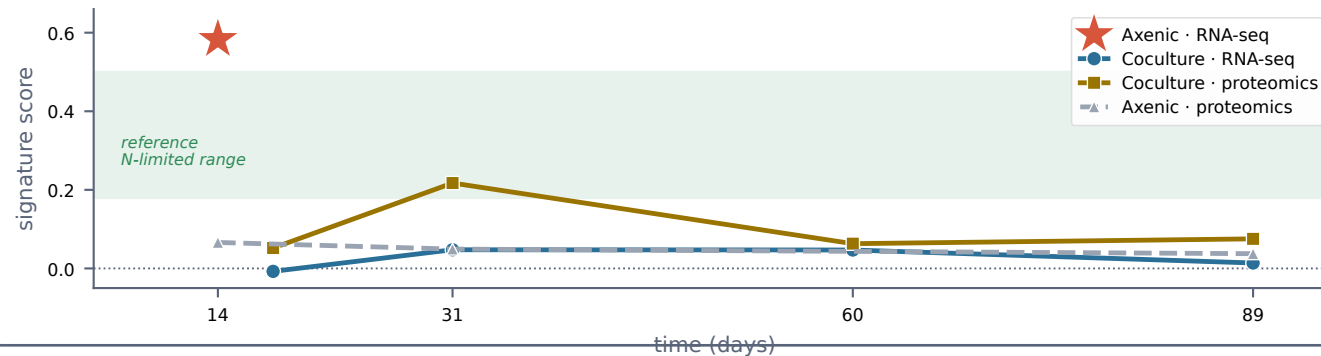
survives 90+ days

Same genotype, same medium — a heterotrophic partner is the only difference.

The analysis, built on the KG

- Pulled RNA-seq + proteomics for MED4 in both conditions
- Built a 189-gene nitrogen-limitation signature from two independent reference studies
- Scored each condition and compared RNA vs protein on genes seen by both

Result: 189-gene N-stress signature score over the 90-day course



The finding

Coculture abolishes the transcriptional nitrogen-stress response, yet a protein-level signature persists — a genuine RNA/protein discordance (not a coverage artifact).

The KG + agent reproduced the published rescue phenotype and surfaced a new post-transcriptional layer — in days, scripted and reproducible, with caveats attached.